UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	OCTOBER 2026
TIME	2 HOURS

NSTRUCTIONS TO CANDIDATES

1. This question paper consists of 5 questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) a one – page Appendix 1
   iii) an Answer Booklet - provided by the Faculty
   iv) a graph paper - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages

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QUESTION 1

a) Solve the inequality $-2x^2 + 4x + 16 > 0$.

(5 marks)



Figure: critical values.

b) Solve the inequality $|-5x - 3| < 3$. Express the answer in interval notation and on a number line.

(5 marks)



Figure: solution on a number line.

c) Sketch $y = 2|x - 4| - 4$. State the vertex, the y-intercept and any x-intercepts.

(5 marks)

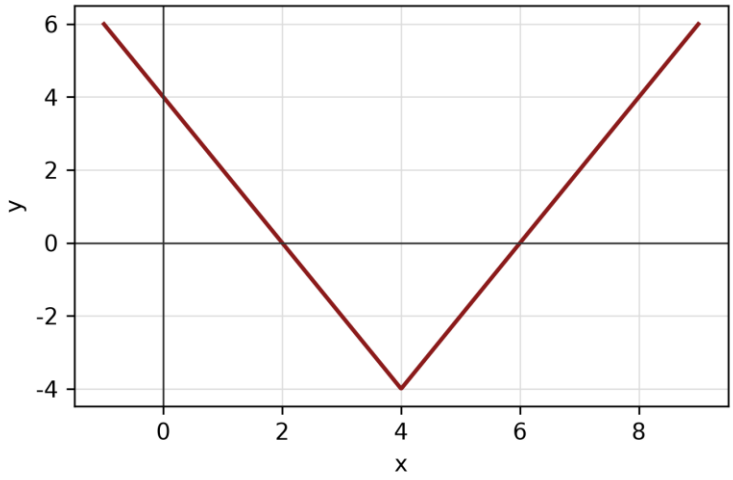


Figure: absolute-value graph.

QUESTION 2

a) Given $f(x) = (4x - 2)/(3x + 6)$, $x \neq -2$, find $f^{-1}(x)$ and state its domain.

(5 marks)

b) The piecewise function is defined by
$f(x) = 4x + 1$, if $x < -1$;
$f(x) = 1$, if $-1 \le x \le 2$;
$f(x) = x^2 + 4$, if $x > 2$.
Sketch the graph and find f(-2), f(0) and f(3).

(5 marks)

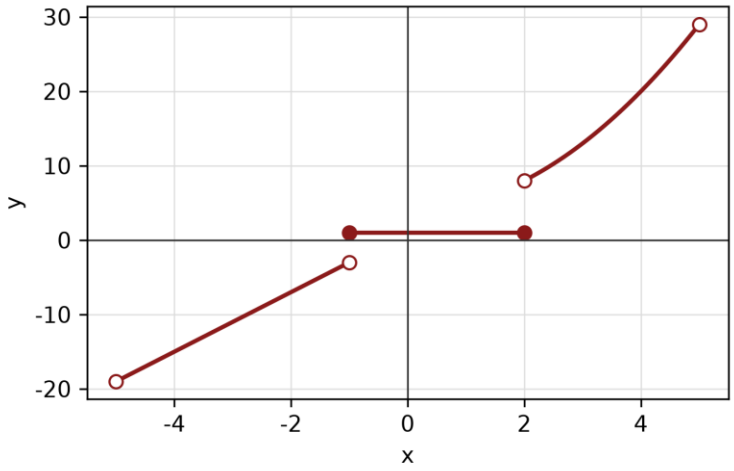


Figure: piecewise graph.

c) The endpoints of a diameter of a circle are (-1, -5) and (-6, -3). Find the centre, radius and equation of the circle.

(5 marks)

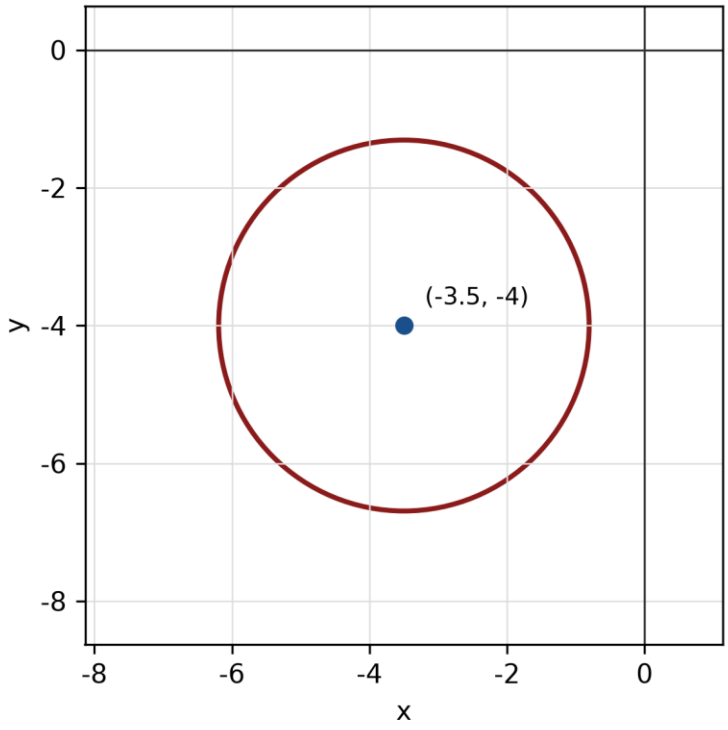


Figure: circle from diameter.

QUESTION 3

a) Given $z_1 = -4 + 5i$ and $z_2 = -1 - 2i$, express z_1/z_2 in the form a + bi.

(5 marks)

b) Find the modulus and argument of $z = 5 + 12i$. Give the argument in degrees correct to one

decimal place.

(5 marks)

c) Solve the logarithmic equation log_2(x) + log_2(x - 31) = 5.

(5 marks)

QUESTION 4

a) Graph the system of inequalities below and shade the feasible region. State the vertices of the region.
$x \geq 0, y \geq 0, 3x + 3y \leq 8$.

(5 marks)

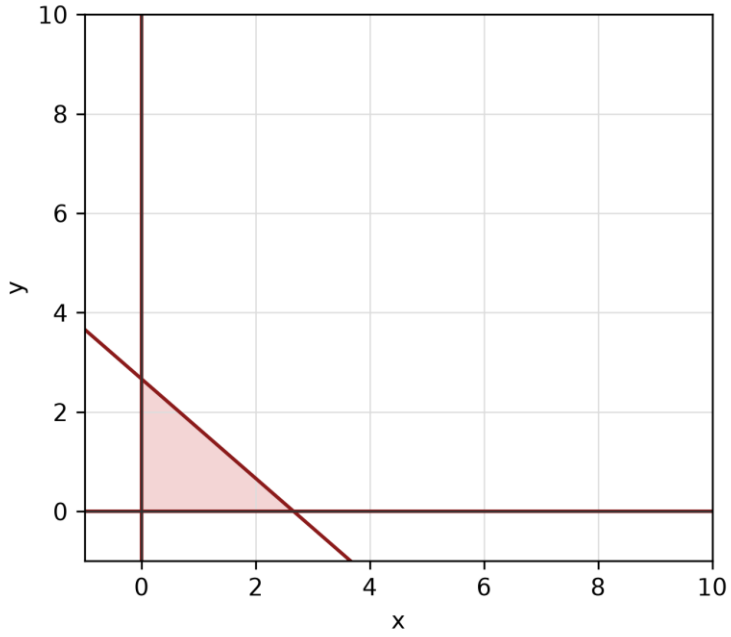


Figure: feasible region.

b) Solve the simultaneous equations
$-x + 3y = 0$
$-3x - y = -10$.

(4 marks)

c) Write $y = x^2/8 + x/4 - 15/8$ in standard form. Hence state the vertex, focus and directrix, and sketch the parabola.

(5 marks)

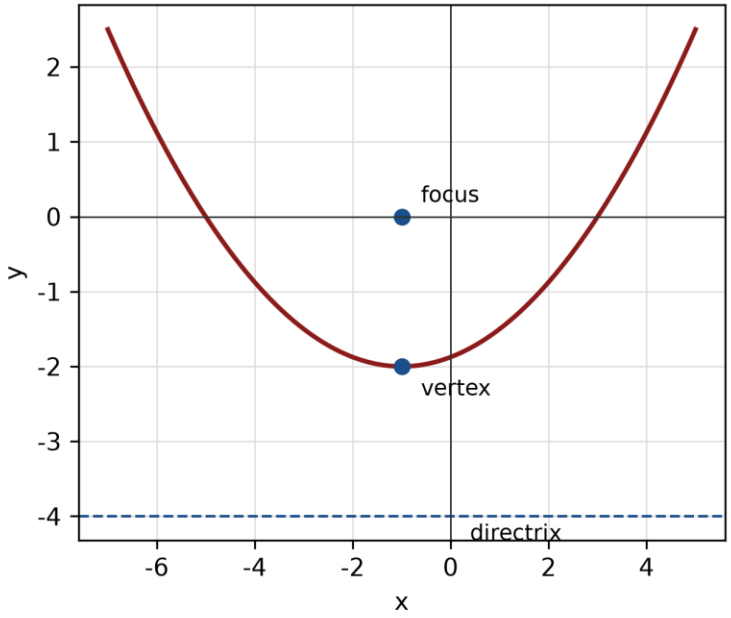


Figure: parabola.

QUESTION 5

a) Prove the identity $\sin^2 x/(1 - \cos x) = 1 + \cos x$.

(5 marks)

b) A circle has radius 10 cm. Find the length of the arc subtended by a central angle of 135°.

(4 marks)

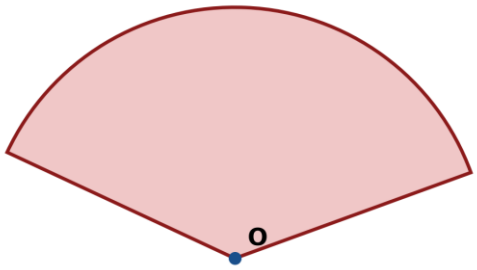


Figure: arc and sector.

c) Find the area of triangle ABC with a = 7 cm, b = 6 cm and included angle C = 30°.

(4 marks)

END OF QUESTION PAPER

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APPENDIX 1

Formula sheet (as supplied with the paper).

$\sin^2 A + \cos^2 A = 1$
$\sin(A + B) = \sin A \cos B + \cos A \sin B$
$\cos(A + B) = \cos A \cos B - \sin A \sin B$
$\tan(A - B) = (\tan A - \tan B) / (1 + \tan A \tan B)$
$\cos 2A = \cos^2 A - \sin^2 A$
$\cos 2A = 1 - 2\sin^2 A = 2\cos^2 A - 1$
$\sin 2A = 2 \sin A \cos A$
$\tan^2 A + 1 = \sec^2 A$
$\tan 2A = 2 \tan A / (1 - \tan^2 A)$
Arc length, $s = r\theta$
Area of sector = $\frac{1}{2}r^2\theta$
$a / \sin A = b / \sin B = c / \sin C$
$a^2 = b^2 + c^2 - 2bc \cos A$
Area of triangle = $\frac{1}{2}ab \sin C$

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